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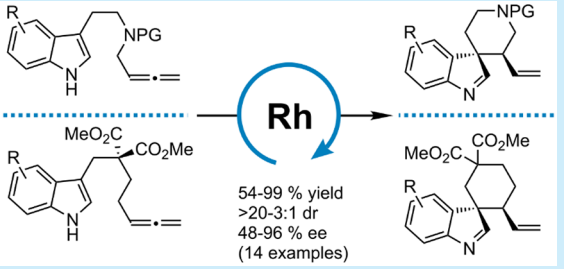
# Rhodium-Catalyzed Stereoselective Cyclization of 3-Allenylindoles and *N*-Allenyltryptamines to Functionalized Vinylic Spiroindolenines

Antonia Becker, Christian P. Grugel, and Bernhard Breit*

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**ABSTRACT:** Herein, we report a highly enantio- and diastereoselective rhodium-catalyzed cyclization of *N*-allenyltryptamines and 3-allenylindoles to 6-membered spirocyclic indolenines. This allylic addition methodology offers the advantage of using a comparably cheap commercially available ligand with low loadings of an affordable rhodium precursor. The products can be converted into functionalized spirooxindoles and spiroindolines, which are regarded as important building blocks for the synthesis of a lot of natural products with biological activities.




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With more than 1000 known compounds, indole alkaloids represent an essential group of natural products. Many of these natural products exhibit a wide range of biological activities. A shared feature of many of these alkaloids is the amine moiety with a C2-linker attached to the indole 3-position, derived from the naturally occurring amino acid L-tryptophane.[1]

In this work, specifically the molecular scaffold of spirocyclic indolenines and their closely related structures are relevant. Three prominent representatives thereof are shown in Figure 1. Stryvomicine shows neuroprotective activity, and perophoramidine acts cytotoxic against specific cancer cells.[2] Koumine is an alkaloid of *Gelsemium elegans Benth.*, which shows anti-inflammatory properties.[3] Given their ability to be easily oxidized at the imine moiety, spirocyclic indolenines allow for a convenient synthesis of spirooxindoles. This gives access to an entire group of compounds with a plethora of pharmacological effects.[4]

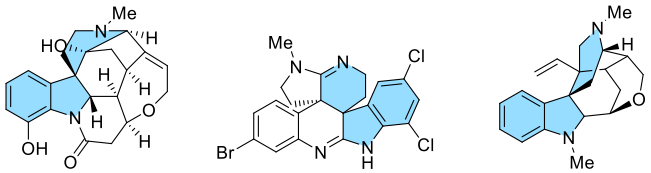



**Figure 1.** Selected biologically active natural products containing a spiroindoline core.

To construct these valuable building blocks in an efficient and stereoselective way, new synthetic methodologies need to be established. An overview on the synthetic developments thus far, regarding the topic of spirocyclic indolenines is outlined in a recent review by Zheng and You.[5] Especially noteworthy concerning the field of intramolecular, stereoselective formation of spiroindolenines are the contributions by the You group. Using chiral phosphoramidite ligands, they developed an Ir-catalyzed asymmetric allylic substitution of indole-3-yl allylic carbonates with a broad scope.[6−9] There are also several accounts of Au-, Ag-, or Cu-catalyzed procedures using tethered terminal or internal alkynes, which lead to the aforementioned products in a stereoselective manner.[10]

In recent years, we reported a series of rhodium-catalyzed chemo-, regio-, and stereoselective allylic addition reactions, coupling allenes, as well as alkynes as allylic precursors with several different pronucleophiles.[11] These hydrofunctionalizations are useful tools to form new branch-selective C−heteroatom[12] and C−C[13] bonds. Especially the selective formation of C−C bonds by allylic addition still represents an ongoing challenge for organic synthesis. In this respect, we recently reported a rhodium-catalyzed enantioselective cyclization of 3-allenylindoles to functionalized tetrahydrocarbazoles (Scheme 1A).[14] Following the confirmation that this cyclization proceeds via a transient spirocyclic intermediate as previously described by the You group,[7] the corresponding spiroindolines could be obtained via in situ reduction with Hantzsch ester.[15]

These findings lead to the work portrayed in this article. Compared to the aforementioned 5-membered spirocycles, the homologue 6-membered spirocyclic indolenines are less prone to isomerization because of their lower ring strain and thus,

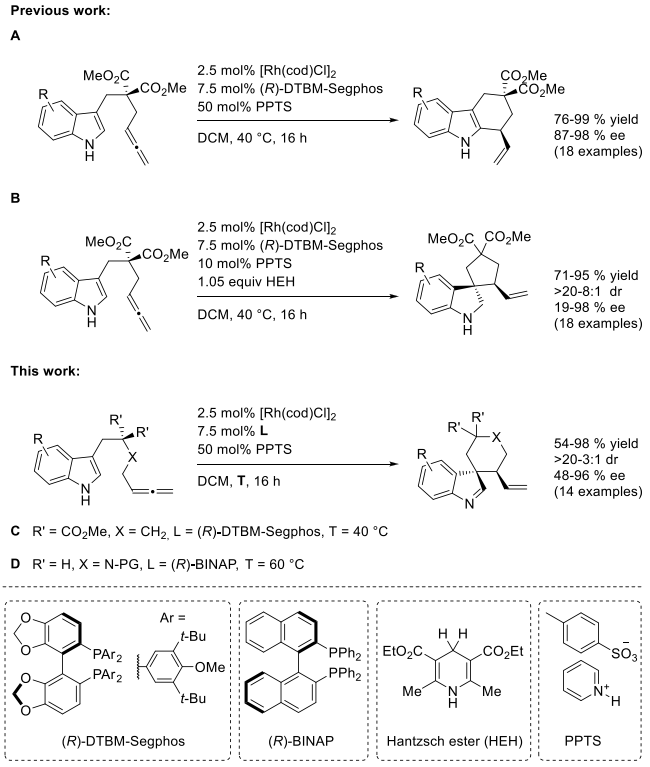
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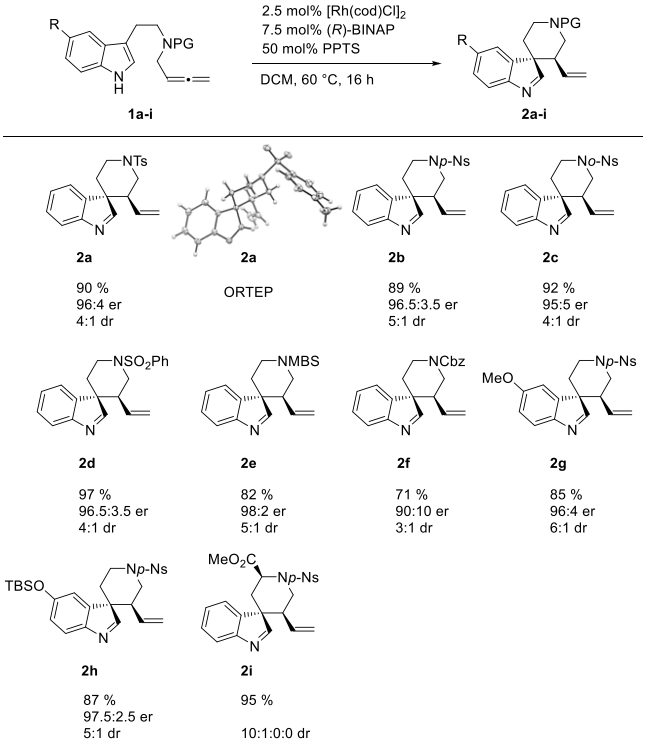
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**Scheme 1. Previous Work by the Breit Group**




**Scheme 2. Reaction Scope of Tryptamine Derivatives**[a]




[a]All yields are isolated yields. *er* was determined by chiral HPLC, and *dr* was determined by $^1$H NMR analysis of the crude product. The absolute configuration of the products was determined by crystal structure analysis of **2a** and assigned by analogy to all other compounds. ORTEP structure with 50% ellipsoid probability.

higher intrinsic stability. This should allow for their selective synthesis and would be an improvement regarding atom economy compared to previous protocols using allylic substitution.[6,16] Since many natural products are based on the abundantly present precursor tryptamine, we decided to incorporate this in the model substrate.

Subjecting the tosyl-protected model substrate **1a** to the synthetic protocol from the tetrahydrocarbazole synthesis (see Scheme 1A) gave the desired product, although the yield and stereoselectivity were moderate after 24 h (70%, 82:18 er, 2:1 dr). Therefore, further optimization of the ligand toward optimal reaction conditions for tryptamine-containing substrates was carried out. Because of the extent of preceding optimizations, it was refrained from testing different classes of privileged ligands, but rather use axial chiral ligands derived from the BINAP and related scaffolds. Both, (*R*)-Segphos and (*R*)-BINAP gave excellent selectivities and the previously prolonged reaction time could be reduced to 16 h overnight reaction with increased temperature. The optimized reaction conditions are shown in Scheme 2.[17]

The scope of the *N*-allenyltryptamines was extended by applying different protecting groups as well as substituents in 5-position of the indole core as shown in Scheme 2. All aromatic sulfonyl amide protecting groups gave very good results with yields of up to 97% and enantiomeric ratios of over 95:5, though the diastereoselectivity remained mediocre. The carbamate protection group Cbz gave slightly inferior results, whereas several oxygen-based functional groups in 5-position were well-tolerated. Additionally, the tryptophane-based substrate **1i** was prepared and successfully converted into the corresponding spirocyclic indolenine with excellent yield and a satisfying diastereoselectivity.

The relative configuration of the cyclization products **2** was first determined by $^1$H NMR comparison with literature data of compound **2a** and assigned to all other derivatives by analogy.[16] The absolute configuration was determined only after enantiomerically pure crystals of compound **2a** were obtained. The acquired X-ray crystal structure is depicted in Scheme 2 as ORTEP representation and corresponds to the (*S*,*S*)-enantiomer.

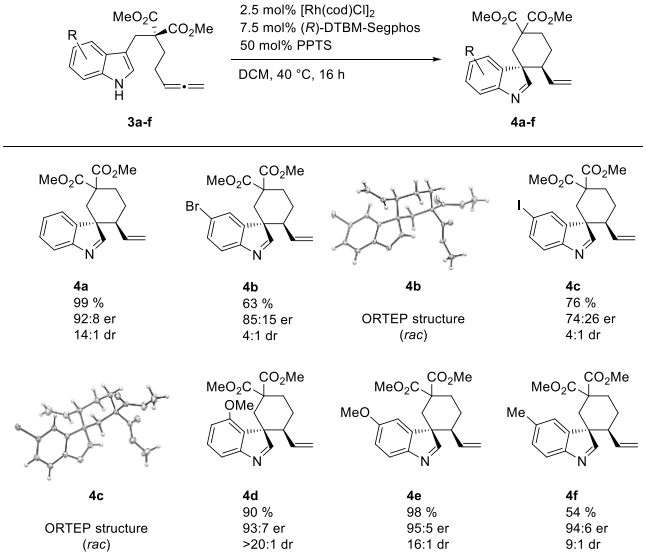
Reactions with model substrate **3a** derived from the preceding investigations leading to 5-membered spirocyclic products, gave good results under the catalytic conditions previously published, as shown in Scheme 3. Applying a lower temperature results in an increased enantioselectivity but is accompanied by a slight drop in yield (30 °C, 70% yield, 95:5 *er*, 13:1 *dr*).

For a small scope of this 3-allenylindole cyclization, a few selected functional groups were introduced to the indole core. All of the applied derivatives **3a**−**f** showed a satisfactory outcome under the shown catalytic conditions (Scheme 3). In case of 5-bromo- and 5-iodo-substituted spiroindolenines (**4b** and **4c**) centrosymmetric crystal structures could be obtained, which are further proof of the observed diastereoselectivity.

In analogy to the postulated catalytic cycle for the 5-membered spirocycle and tetrahydrocarbazole formation, we would expect this transformation to proceed via formation of a rhodium(III) hydride complex with PPTS, hydrometalation of the allene to give an allyl-rhodium(III) intermediate, which then undergoes a nucleophilic attack by the indole 3-position.[14] Interestingly, formation of the 6-membered spirocycle results in an inverted diastereoselectivity compared to the 5-membered ring formation. The differences of the transition state (TS) geometry seems to be a possible rationale: While


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**Scheme 3. Reaction Scope of Malonate-Tethered Vinylic Spiroindolenines**[a]




[a]All yields are isolated yields. er was determined by chiral HPLC, dr was determined by $^1$H NMR analysis of the crude product. The absolute configuration of the products was assigned by analogy to **2a**. ORTEP structures with 50% ellipsoid probability.

the 6-membered ring formation might occur through a chairlike TS, the 5-membered spirocycle likely is formed via an envelope transition state geometry, each favoring a pseudoequatorial orientation of the allyl rhodium fragment (Figure 2).

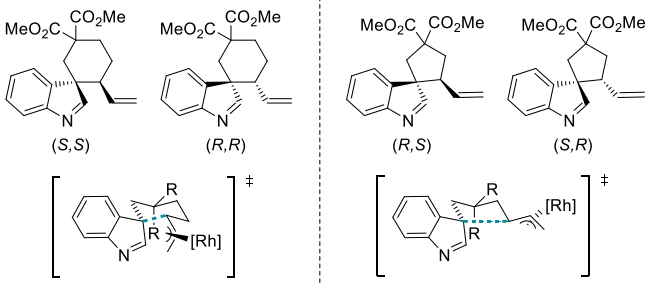
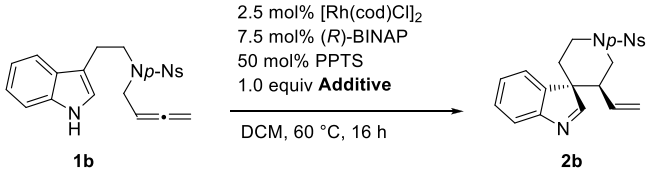



**Figure 2.** Postulated transition states for 5-membered versus 6-membered spirocycle formation as possible explanation for the observed diastereoselectivity.

To show the robustness of this transformation, several experiments with carefully chosen additives have been performed. This selection is comprised of competing nucleophiles and electrophiles, each bearing potentially reactive functional groups. Moreover, the reaction was carried out without precaution to moisture and even with the addition of water. The results are listed in Table 1.

The competing nucleophiles benzaldehyde, benzoic acid, ethanol and phenol (entries 1, 2, 4, and 5) could be added to the reaction mixture without significant differences in the reaction outcome. When using one of the best nucleophiles in allylic additions, 4-bromopyrazole, yield and enantioselectivity of **2b** dropped (entry 3). Nevertheless, the intramolecular reaction is still somewhat favored over an intermolecular attack. Addition of a competing terminal allene (entry 6)

**Table 1. Robustness Screening**




| entry | additive | yield[a] (%) | dr[b] | er[c] |
|---|---|---|---|---|
| 1 | benzaldehyde | 88 | 7.5:1 | 98.5:1.5 |
| 2 | benzoic acid | 83 | 5:1 | 93:7 |
| 3 | 4-bromopyrazole | 62 | 6:1 | 75:25 |
| 4 | ethanol | 83 | 7.5:1 | 96.5:3.5 |
| 5 | phenol | 84 | 6.5:1 | 97:3 |
| 6 | penta-3,4-dien-1-ylbenzene | 29 | 6.5:1 | 97:3 |
| 7[d] | prop-1-yn-1-ylbenzene | 83 | 6:1 | 97:3 |
| 8 | 2-TMS-buta-1,3-diene | 71 | 5:1 | 95:5 |
| 9 | (non flame-dried flask) | 84 | 9:1 | 95.5:4.5 |
| 10 | water | 81 | 6:1 | 96:4 |

[a]Isolated yield. [b]Determined by 1H NMR analysis of the isolated compound. [c]Determined by chiral HPLC. [d]70% unreacted alkyne determined via crude $^1$H NMR.

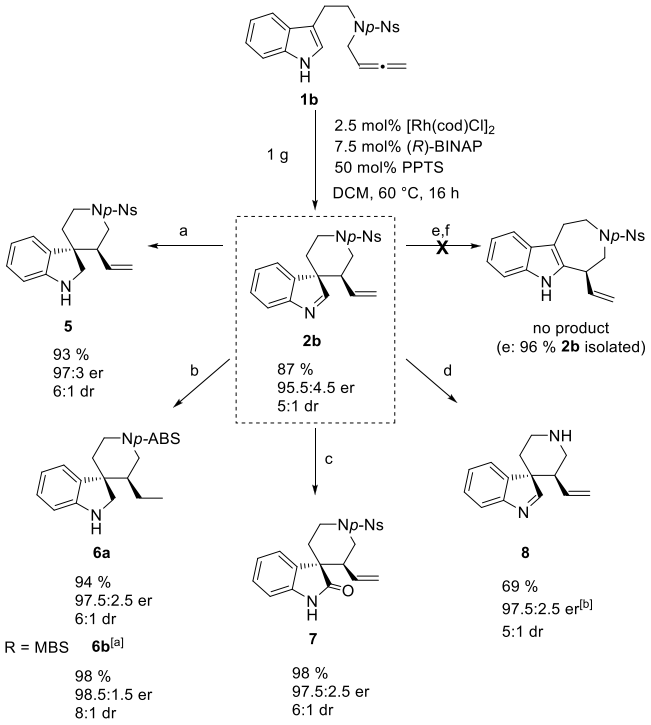
resulted in a lower yield. It could be shown that an internal alkyne, a known intermediate in rhodium-catalyzed allylic additions, cannot compete with the intramolecular C−C-bond formation (see entry 7).[11] Addition of 2-TMS-buta-1,3-diene resulted in only a slight decrease in the yield. Therefore, a coordination to the catalyst resulting in a blockage thereof could be ruled out. As previously shown for the formation of tetrahydrocarbazoles, this system is highly insensitive when it comes to moisture.[14] The reaction could be carried out in a non-flame-dried flask and even the addition of one equivalent of water did not significantly influence the reaction outcome.

The scalability of this methodology was demonstrated by performing a reaction at a 1 gram scale with substrate **1b**, achieving a comparable yield and stereoselectivity. The obtained spirocyclic indolenine undergoes versatile transformations, as shown in Scheme 4. Selective reduction of the imine moiety was accomplished by the use of sodium cyanoborohydride in methanol, while harsher reaction conditions lead to reduction of imine and terminal alkene. When using the *p*-nosyl protecting group, the nitro-group was reduced to the corresponding amine, which could be avoided with the MBS protecting group. The corresponding oxindole **7** was obtained after selective oxidation using Pinnick oxidation conditions.[18] Furthermore, reaction with *p*-thiocresole and potassium carbonate in DMF gave the deprotected spiroindolenine **8**.[19] As expected for this ring size, application of the conditions for acid-catalyzed migration did not lead to an annulated 7-membered heterocycle.[8] In all of the above-mentioned transformations no loss of optical purity was observed.

In conclusion, we herein present a method to convert 3-tethered allenyl indoles **1** and **3** to 6-membered spirocyclic indolenines **2** and **4** via rhodium-catalyzed allylic addition, giving access to valuable products with high stereoselectivities. Aside from the general advantage of using allylic addition compared to substitution reactions regarding the higher atom economy, the low loadings of commercially available, affordable ligand and the robustness of the reaction are benefits of this protocol. The versatility was demonstrated by


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**Scheme 4. Gram-Scale Catalysis and Selected Transformations of Spiroindolenine 2b**[z]




[z]All yields given as isolated yields. *dr* was determined by $^1$H NMR analysis of the isolated product. *er* was determined by chiral HPLC. Reactions and conditions: (a) $NaBH_3CN$ (1.0 equiv), MeOH, 0 °C to rt, 16 h; (b) Pd/C (5 wt %, 10 mol % Pd), $H_2$ (1 atm, via balloon), MeOH, rt, 16 h, (ABS = 4-aminobenzenesulfonyl); [a] was synthesized from **2e**; (c) 2-methyl-2-butene (10 equiv), $NaClO_2$ (5.0 equiv), $NaH_2PO_4$ (aq. 1 M, 1.5 equiv), THF, rt, 7 h; (d) *p*-thiocresole (1.5 equiv), $K_2CO_3$ (3.0 equiv), DMF, 50 °C, 16 h; (b) *er* determined after conversion to derivative **2a**; (e) TsOH (30 mol %), THF, rt to 60 °C, 4 h; (f) HCl (g) in THF (4.5 M), rt, 16 h.

the possibility to use malonate tethered substrates as well as tryptamine derivatives with a broad functional group tolerance. Additionally noteworthy is the potential to transform the cyclization products to valuable building blocks for natural product synthesis, such as functionalized spirooxindoles and spiroindolines.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c01234.

Experimental details and analytical data for the synthesized compounds (PDF)

### Accession Codes

CCDC 1956390, 1966016, and 2008562 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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### Notes

The authors declare no competing financial interest.

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